# Survival Analysis II: Applications to Mortality Data

## Survival Estimation, Cox Regression, and Nonlinear Associations

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# Outline


1. **Review of Survival Analysis Methods**
   - Kaplan–Meier and Nelson–Aalen estimators
   - Cox proportional hazards model
2. **NHANES Mortality Study**
   - Study population and variables
3. **Survival Analysis Results**
   - Survival curves and Cox regression
4. **Nonlinear Associations**
   - Restricted cubic spline Cox regression
   - Effect modification
5. **Discussion and Conclusions**

# Kaplan–Meier Estimator: A Review

**Survival function:**

$$S(t) = P(T^* > t)$$

**Kaplan–Meier estimator:**

$$\widehat{S}(t) = \prod_{t_j \leq t} \left(1 - \frac{d_j}{Y(t_j)}\right)$$

## Implementation in R

```
library(survival)
km_fit <- survfit(Surv(time, status) ~ group, data = dataset)
plot(km_fit)
```

- Estimates survival probabilities over time.
- Allows comparison of survival across groups.

# Nelson–Aalen Estimator: A Review

**Cumulative hazard function:**

$$A(t) = \int_0^t \alpha(s)\, ds$$

**Nelson–Aalen estimator:**

$$\widehat{A}(t) = \sum_{t_j \le t} \frac{d_j}{Y(t_j)}$$

## Implementation in R

```
library(survival)
na_fit <- survfit(Surv(time, status) ~ group, data = dataset,
 type = "fleming-harrington")
plot(na_fit, fun = "cumhaz")
```

- Estimates cumulative hazard over time.
- Allows comparison of cumulative hazards across groups.

# Cox Proportional Hazards Model: A Review

**Cox proportional hazards model:**

$$\alpha(t \mid \mathbf{x}) = \alpha_0(t) \exp\left(\boldsymbol{\beta}^\top \mathbf{x}\right)$$

- $\alpha_0(t)$: unspecified baseline hazard.
- $\boldsymbol{\beta}$: regression coefficients.
- $\exp(\beta_j)$: hazard ratio for a one-unit increase in $x_j$, holding other covariates fixed.

## Implementation in R

```
cox_fit <- coxph(
  Surv(time, status) ~ x1 + x2,
  data = dataset
)
summary(cox_fit)
```

**Key assumption:** Hazard ratios remain constant over time.

# Research Motivation

## Research Question

How is glycated hemoglobin (HbA1c) associated with all-cause mortality among U.S. adults?

- HbA1c reflects average blood glucose levels over approximately 2–3 months.
- Both low and high HbA1c levels may be associated with increased mortality risk.
- The HbA1c–mortality association may be **nonlinear**.
- Associations may differ across demographic and clinical subgroups.

# NHANES: Data Sources and Availability

**National Health and Nutrition Examination Survey (NHANES)**

- Conducted by the **National Center for Health Statistics (NCHS)**, part of the CDC.
- Nationally representative survey of the U.S. civilian, noninstitutionalized population.
- Public-use data available in two-year survey cycles, including 1999–2018.

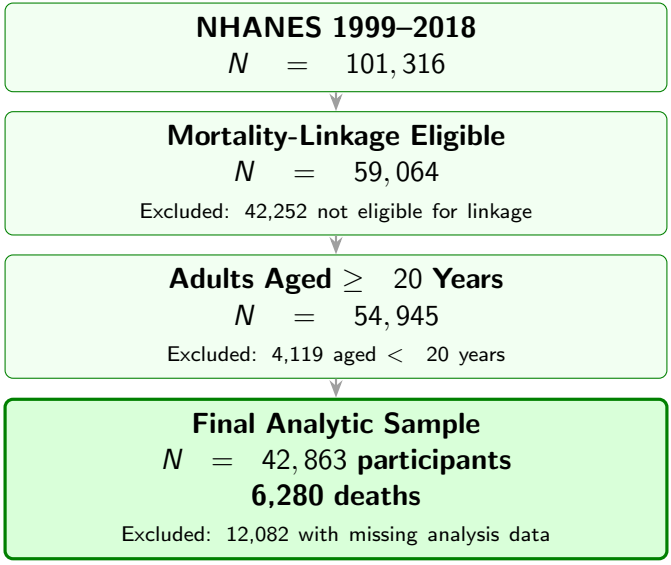
## Major Data Components

| | |
|---|---|
| **Demographics** | Age, sex, race/ethnicity, education, income |
| **Questionnaires** | Smoking, diabetes, hypertension, health history |
| **Examinations** | Blood pressure, height, weight, BMI |
| **Laboratory** | HbA1c, cholesterol, glucose, blood biomarkers |
| **Mortality Linkage** | Vital status, follow-up time, cause of death |

## Why NHANES for Survival Analysis?

NHANES combines detailed health measurements with **long-term mortality follow-up**, allowing researchers to examine associations between risk factors and mortality.

# NHANES Data and Study Population




Mortality linked through December 31, 2019.

# Study Variables and Survival Outcome

**Survival Outcome**

- **Follow-up Time:** Years from NHANES examination to death or censoring.
- **Event Indicator:** All-cause mortality ($\delta = 1$); otherwise right-censored ($\delta = 0$).

**Main Predictor: HbA1c**

| Category | HbA1c Level |
|---|---|
| Normal range | $< 5.7\%$ |
| Prediabetes range | 5.7%–6.4% |
| Diabetes range | $\geq 6.5\%$ |

## Adjustment Variables

Age, sex, race/ethnicity, education, income, BMI, smoking status, hypertension, diagnosed diabetes, and total cholesterol.

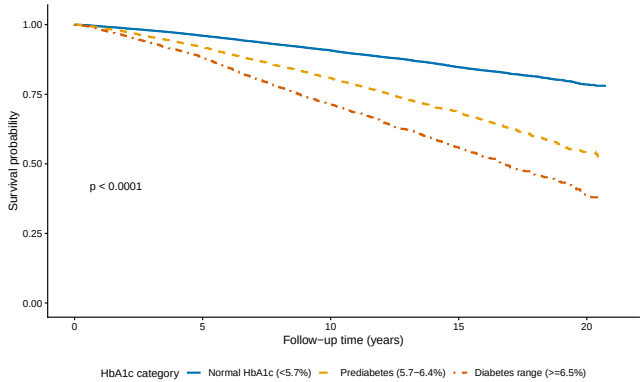
# Statistical Analysis Overview

1. **Kaplan–Meier Estimator**
   - Estimate survival probabilities by HbA1c group.
2. **Nelson–Aalen Estimator**
   - Estimate cumulative hazards by HbA1c group.
3. **Survey-Weighted Cox Regression**
   - Estimate adjusted hazard ratios for mortality.
4. **Restricted Cubic Spline Cox Regression**
   - Examine nonlinear HbA1c–mortality associations.

## Additional Analysis

Evaluate effect modification by age, smoking status, and hypertension.

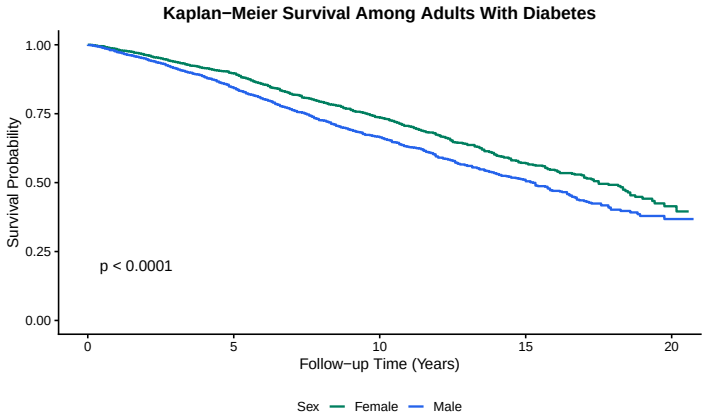
# Kaplan–Meier Survival by HbA1c Group




## Key Finding

Survival differed across HbA1c categories (**log-rank** $p < 0.001$). Participants in the diabetes-range group had the **lowest observed survival**.

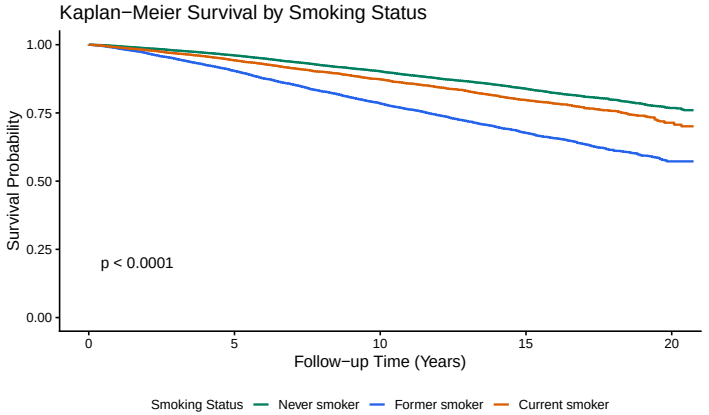
# Survival by Sex Among Adults With Diabetes




## Key Finding

Among adults with diabetes, males had **lower observed survival** than females (**log-rank** $\chi^2 = 25.42$**,** $p < 0.001$).

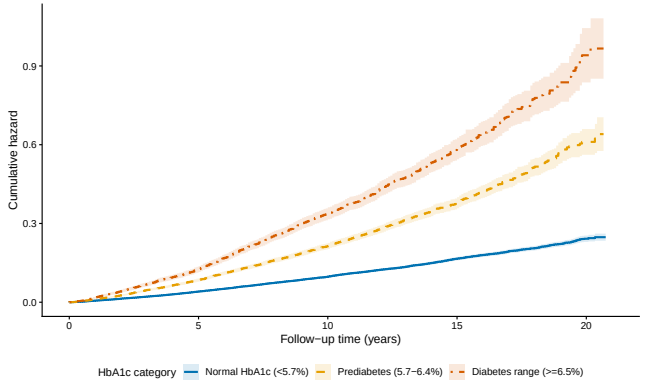
# Kaplan–Meier Survival by Smoking Status




## Key Finding

Survival differed significantly across smoking groups (**log-rank** $p < 0.001$).
Never smokers had the highest observed survival, while former smokers had the lowest. These are **unadjusted comparisons**.

# Cumulative Hazard by HbA1c Group




## Key Findings

- The diabetes-range group had the **highest estimated cumulative hazard**.
- The prediabetes group showed an intermediate cumulative hazard.
- The differences in cumulative hazard between groups became more pronounced over follow-up.

# Why Do We Need Cox Regression?

## Limitation of Unadjusted Comparisons

Kaplan–Meier and Nelson–Aalen estimators compare survival experiences without adjusting for other risk factors.

- HbA1c groups may differ in **age, sex, smoking status**, and other characteristics.
- These characteristics may also be associated with mortality risk.
- Cox regression allows us to estimate the **HbA1c–mortality association** while adjusting for other covariates.

## Research Question

Is HbA1c associated with all-cause mortality **after adjusting for demographic and clinical risk factors**?

# Why Survey-Weighted Cox Regression?

**NHANES uses a complex, multistage probability sampling design rather than simple random sampling.**

- **Sampling Weights:** Account for unequal selection probabilities and survey nonresponse.
- **Stratification:** Reflects the division of the population into sampling strata.
- **Clustering:** Accounts for dependence among participants within primary sampling units (PSUs).

## Why Does This Matter?

Survey-weighted Cox regression incorporates the NHANES sampling design to obtain population-relevant hazard ratio estimates and design-adjusted standard errors.

# Multivariable Cox Regression Results

**Survey-weighted Cox proportional hazards model**

| Risk Factor | Adjusted HR | 95% CI |
|---|---|---|
| HbA1c (per 1% increase) | 1.08 | (1.04, 1.12) |
| Age (per year) | 1.10 | (1.09, 1.10) |
| Male (vs. female) | 1.44 | (1.36, 1.53) |
| Current smoker (vs. never) | 2.38 | (2.16, 2.61) |
| Hypertension (yes vs. no) | 1.28 | (1.19, 1.38) |
| Diabetes (yes vs. no) | 1.43 | (1.30, 1.57) |

## Key Findings

- After adjusting for demographic and clinical characteristics, HbA1c remained **significantly associated with mortality**.
- Each 1-percentage-point increase in HbA1c was associated with an **8% higher hazard of all-cause mortality**.
- Adjusted HR = 1.08 (95% CI: 1.04–1.12).

*HR = hazard ratio; CI = confidence interval. Selected covariates shown.*

# Why Examine Nonlinear Associations?

## Finding from the Cox Regression Model

Each 1-percentage-point increase in HbA1c was associated with an estimated **8% higher mortality hazard** (HR = 1.08; 95% CI: 1.04–1.12).

**But does this association remain the same across all HbA1c levels?**

- The standard Cox model assumes a **linear association with the log-hazard**.
- Mortality risk may increase at both **low and high HbA1c levels**.
- A single hazard ratio may not adequately describe such a relationship.

## Our Approach

Use **restricted cubic spline Cox regression** to examine the shape of the HbA1c–mortality association.

# Extending Cox Regression to Nonlinear Associations

**Standard Cox model:**

$$\alpha(t \mid x, \mathbf{z}) = \alpha_0(t) \exp\left(\beta x + \boldsymbol{\gamma}^\top \mathbf{z}\right)$$

- Assumes a linear effect of $x$ on the log-hazard.

**Cox model with restricted cubic splines:**

$$\alpha(t \mid x, \mathbf{z}) = \alpha_0(t) \exp\left(f(x) + \boldsymbol{\gamma}^\top \mathbf{z}\right)$$

- $x$: continuous predictor of interest; $\mathbf{z}$: vector of other covariates.
- $f(x)$: smooth function allowing a nonlinear association between $x$ and the log-hazard.

## Implementation in R

```
library(rms)

spline_fit <- cph(
  Surv(time, status) ~ rcs(x, 4) + z1 + z2,
  data = dataset
)
```

# Restricted Cubic Splines in Our NHANES Study

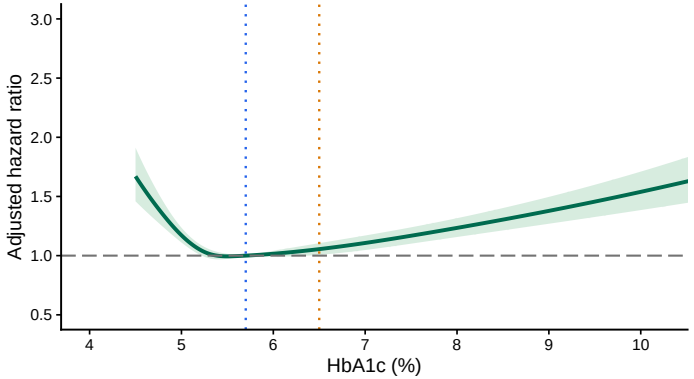
**Application:** HbA1c and all-cause mortality among U.S. adults.

- **Predictor:** HbA1c modeled continuously.
- **Spline Specification:** Four knots at HbA1c values of 4.8%, 5.3%, 5.6%, and 7.6%.
- **Reference Value:** HbA1c $= 5.7\%$ (HR $= 1$).
- **Covariate Adjustment:** Demographic and clinical risk factors.
- **Statistical Inference:** Test for evidence of nonlinearity.

**Interpretation**

The fitted spline provides adjusted hazard ratios across HbA1c levels relative to the reference value of 5.7%.

# Nonlinear Association Between HbA1c and Mortality




## Key Findings

- The association between HbA1c and mortality was **nonlinear** ($p < 0.001$).
- The estimated mortality hazard was lowest near **HbA1c = 5.5%**.
- Higher estimated hazards were observed at both low and high HbA1c levels, suggesting a **U-shaped association**.

# Interpreting the Nonlinear Association

## What Does the U-Shaped Pattern Suggest?

- The association between HbA1c and mortality is **not adequately summarized by a single hazard ratio**.
- The association varies across the HbA1c range, highlighting the value of **flexible regression models**.

## Important Considerations

- Elevated mortality hazards at low HbA1c levels may reflect underlying illness, reverse causation, or residual confounding.
- The estimated minimum is **not a recommended clinical target**.
- These observational associations **do not establish causation**.

# Effect Modification of the HbA1c–Mortality Association

## Research Question

Does the association between HbA1c and mortality differ across demographic and clinical subgroups?

**Potential Effect Modifiers**

- **Age:** Does the association differ between younger and older adults?
- **Smoking Status:** Does the association differ across smoking groups?
- **Hypertension:** Does the association differ by hypertension status?

## Statistical Approach

Include interaction terms in the Cox model and test whether the HbA1c–mortality association differs across subgroups.

# Testing Effect Modification in Cox Regression

**Cox model with an interaction term:**

$$\alpha(t \mid x, z, \mathbf{w}) = \alpha_0(t) \exp\left(\beta_1 x + \beta_2 z + \beta_3 xz + \boldsymbol{\gamma}^\top \mathbf{w}\right)$$

- $x$: HbA1c level; $z$: potential effect modifier.
- $\mathbf{w}$: other adjustment covariates.
- $\beta_3$: interaction coefficient.

## Hypothesis Test

$$H_0 : \beta_3 = 0 \qquad \text{vs.} \qquad H_A : \beta_3 \neq 0$$

A significant interaction indicates that the HbA1c association differs across levels of $z$.

**Effect modifiers examined:** Age, smoking status, and hypertension.

# Effect Modification: Interaction Test Results

**Does the HbA1c–mortality association differ across subgroups?**

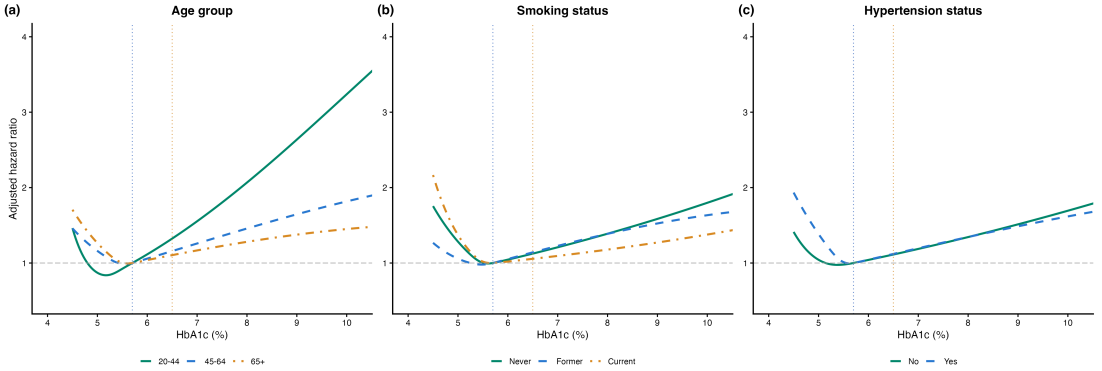
- **Age:** Evidence of effect modification by age.
- **Smoking Status:** Evidence that the association differs across smoking groups.
- **Hypertension:** Evidence that the association differs by hypertension status.

**Interpretation**

The association between HbA1c and all-cause mortality may not be identical across population subgroups.

**Next:** Examine subgroup-specific spline curves to understand how the association differs.

# Effect Modification: Subgroup-Specific Associations




## Key Finding

The subgroup-specific spline curves illustrate differences in the **shape and magnitude of the HbA1c–mortality association** across age, smoking, and hypertension subgroups, relative to HbA1c = 5.7%.

# Interpreting Effect Modification Results

## Age Groups

- **Low HbA1c ($< 5.7\%$):** Adults aged 65+ generally showed the highest relative hazard, while those aged 20–44 showed the lowest near the curve minima.
- **High HbA1c ($\geq 6.5\%$):** Adults aged 20–44 showed the **steepest increase in relative hazard**, followed by ages 45–64 and 65+.

## Smoking Status

- **Low HbA1c:** Current smokers showed the highest estimated relative hazard at the low end.
- **High HbA1c:** Never smokers showed the **largest increase in relative hazard**, while current smokers showed the smallest.

## Hypertension Status

- **Low HbA1c:** Participants with hypertension showed higher estimated relative hazard ratios.

# Study Strengths and Limitations

## Strengths

- Large NHANES sample with **long-term mortality follow-up**.
- Nationally representative survey data with detailed demographic and clinical measures.
- Use of **survey-weighted Cox regression** to account for the complex sampling design.
- Restricted cubic splines allowed **flexible assessment of nonlinear associations**.

## Limitations

- Observational design: **causal conclusions cannot be established**.
- Potential residual confounding and reverse causation.
- HbA1c and other risk factors were measured at baseline; changes during follow-up were not captured.

# Main Conclusions

## Findings from the NHANES Study

- HbA1c was **associated with all-cause mortality** after adjustment for demographic and clinical factors.
- Restricted cubic spline analysis identified a **nonlinear, approximately U-shaped association** between HbA1c and mortality.
- The HbA1c–mortality association showed evidence of **effect modification** by age, smoking status, and hypertension.

## Statistical Takeaways

- Kaplan–Meier and Nelson–Aalen estimators describe survival and cumulative hazard.
- Survey-weighted Cox regression estimates adjusted associations while accounting for the NHANES sampling design.
- Restricted cubic splines and interaction analyses reveal patterns that simpler models may miss.

# Thank You

Questions and Discussion

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